Sina Heidari

Curriculum Vitae

Email:	sinaheidari@vt.edu
Personal Page	https://s1naheidari.github.io
Github:	https://github.com/S1naHeidari
Linkedin:	https://www.linkedin.com/in/sina-heidari-3a935412a/
Google Scholar	https://scholar.google.com/citations?user=-NFnSYgAAAAJ&hl=en&oi=ao

Education

Virginia Tech — Blacksburg, US

PhD — *2024 – present*

Major: Computer Science and Applications

University of Kurdistan — Sanandaj, Iran

Bachelor — *2016 – 2021*

Major: Software Engineering

Research Interests

- High-performance Computing
- Kernel Optimizations
- Agentic AI for Systems
- ML Compilers

Publications

Heterogeneity-aware Load Balancing in Serverless Computing Environments — 2023

Venue: The 7th International Conference on Internet of Things and Its Applications

Status: Published by IEEE

Link: https://ieeexplore.ieee.org/document/10365354

FACT: Compositional Kernel Synthesis with a Three-Stage Agentic Workflow — 2026

Authors: Sina Heidari, Dimitrios S. Nikolopoulos

Venue: arXiv:2604.26666

Link: https://arxiv.org/abs/2604.26666

Certified Kernel Optimization

Status: In preparation

Persistent GPU Kernels for Sequence Models

Status: In preparation

Research Experience

Graduate Research

Performance Engineering for Emerging Architectures Laboratory (PEARL) — *Fall 2024 – present*

Advisor: Prof. Dimitrios S. Nikolopoulos

Research Area: Code generation and optimization for GPU kernels

Graduate Research

Distributed Computing Systems Research Laboratory (DCS Lab) — *Fall 2020 – Fall 2024*

Advisors: Dr. Sadoon Azizi

Research Area: Serverless Computing

Teaching Experience

Virginia Tech

Graduate Teaching Assistant *Spring 2025 – present*
Course title: Computer Systems (4 semesters)

Graduate Teaching Assistant *Fall 2024*
Course Title: Programming Languages

University of Kurdistan

Undergraduate Teaching Assistant *Spring 2018*
Course title: Advanced Programming

Undergraduate Teaching Assistant *Summer 2017 & Fall 2017*
Course Title: Fundamentals of Computer and Programming

Technical Skills

- **Programming and Scripting Languages**
  - Programming: C, C++, Java, C
  - Scripting: Python, bash, and sed
  - Web: CSS, HTML, Django and Java Script
- **GPU Programming and Kernel Optimization**
  - CUDA ecosystem: CUTLASS, CuTe, cuBLAS, cuDNN
  - Memory layout and hierarchy optimization, tiling strategies, kernel fusion
  - Optimization of tensor kernels for machine learning workloads
- **GPU Compilers and Machine Learning Compiler Frameworks**
  - ML compilers: TVM, MLIR
  - Compiler infrastructure: LLVM
  - Program analysis, kernel generation, and optimization pipelines for GPU workloads
- **Agentic AI for HPC Optimization**
  - Kernel generation and optimization
  - Agent-based compiler development
  - Domain-specific agentic memory
  - Enhancement of agentic AI using RAG and tool use
- **Systems and Infrastructure**
  - Linux-based systems: Arch Linux, Manjaro, Ubuntu, Debian, CentOS
  - Virtualization and cloud: Docker, Kubernetes, VirtualBox, Vagrant
  - Automation and monitoring: Ansible, Prometheus

Grants

NAIRR Pilot Start-Up Award: Efficient GPU Systems for Large-Model Serving, Training and Kernel Optimization 2026
Award: 2,000 GPU-hours on SDSC Expanse AI
PI: Prof. Dimitrios S. Nikolopoulos (PEARL Lab, Virginia Tech)
Role: Lead author of the proposal

Awards and Honors

Graduate Assistantship, Virginia Tech 2024 – 2029
Best Paper Award: The International Conference on IoT & Applications, University of Isfahan 2023
Granted, National University Tuition Waiver (including partly necessary expenses) 2016
